Lecture Notes II

Course: Analytical Mechanics

Topic: Euler-Lagrange equations of motion¿

References: H. Goldstein, C. Poole, and J. Safko, *Classical Mechanics*, 3rd ed., Section 1.4., G. F. Torres del Castillo, *An Introduction to Hamiltonian Mechanics*, Birkhäuser, (2018)

We will study the formulation of the Euler-Lagrange equations of motion starting from Newton's second law and D'Alembert's principle of virtual work, analyzing the transition to generalized coordinates and the structure of generalized forces. Furthermore, we will establish the criteria for regular and singular Lagrangians through the Hessian matrix, examine the mathematical framework of velocity-dependent potentials, and derive the necessary and sufficient conditions under which a force admits such a potential. Finally, we will apply this formalism to the electromagnetic Lorentz force, demonstrating its equivalence with Maxwell's equations and constructing its corresponding generalized potential and standard Lagrangian.

Euler-Lagrange equations of motion

The second difficulty presented before is - by making the forces of constraint dissappear by formulating mechanics in a different way, in particular, we will develop the Lagrangian and Hamiltonian formulations of mechanics. In order to do so, we will use a device first thought by James Bernoulli and developed by D'Alembert.

The equation of motion for the α-th particle is given by Newton's second law (assuming an inertial reference frame $\mathscr{I}$)

$$\mathbf{F}_\alpha = \dot{p}_\alpha$$

which is equivalent to

$$F_\alpha - \dot{p}_\alpha = 0$$

so that it is trivial to see that

$$\sum_\alpha (\mathbf{F}_\alpha - \dot{p}_\alpha) \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = 0 \quad \text{for each } i = 1, \dots, n$$

and thus decomposing $\mathbf{F}_\alpha = \mathbf{F}_\alpha^{(a)} + \mathbf{F}_\alpha^{(c)}$ we get

$$\sum_\alpha \left(\mathbf{F}_\alpha^{(a)} - \dot{p}_\alpha \right) \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} + \sum_\alpha \mathbf{F}_\alpha^{(c)} \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = 0 \tag{2}$$

We now restrict to systems for which the second term vanishes, i.e.

$$\sum_\alpha \mathbf{F}_\alpha^{(c)} \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = 0 \quad \forall i$$

which is sometimes known as the principle of virtual work. This way, (2) becomes

$$\sum_\alpha \left(\mathbf{F}_\alpha^{(a)} - \dot{p}_\alpha \right) \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = 0$$

which is often called D'Alembert's principle.

From the transformation equations (1) we have that

$$\dot{\boldsymbol{r}}_\alpha \equiv v_\alpha = \frac{\mathrm{d}\boldsymbol{r}_\alpha}{\mathrm{d}t} = \frac{\partial \boldsymbol{r}_\alpha}{\partial q_j} \frac{\partial q_j}{\partial t} + \frac{\partial \boldsymbol{r}_\alpha}{\partial t} = \frac{\partial \boldsymbol{r}_\alpha}{\partial q_j} \dot{q}_j + \frac{\partial \boldsymbol{r}_\alpha}{\partial t}$$

so that

$$\frac{\partial \dot{\boldsymbol{r}}_\alpha}{\partial \dot{q}_i} = \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} \quad \text{(NOT CANCELLING THE DOTS)} \tag{3}$$

and it is easy to show that

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial}{\partial q_i} = \frac{\partial}{\partial q_i}\frac{d}{dt},$$

thus considering the term $\dot{p}_\alpha = m_\alpha \ddot{\boldsymbol{r}}_\alpha$ we see that

$$\begin{aligned} m_\alpha \ddot{\boldsymbol{r}}_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} &= m_\alpha \frac{\mathrm{d}\dot{\boldsymbol{r}}_\alpha}{\mathrm{d}t} \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\left(m_\alpha \dot{\boldsymbol{r}}_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i}\right) - m_\alpha \dot{\boldsymbol{r}}_\alpha \cdot \frac{\partial \dot{\boldsymbol{r}}_\alpha}{\partial q_i} \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left(m_\alpha \dot{\boldsymbol{r}}_\alpha \cdot \frac{\partial \dot{\boldsymbol{r}}_\alpha}{\partial \dot{q}_i}\right) - \frac{\partial}{\partial q_i}\left(\frac{1}{2} m_\alpha \dot{\boldsymbol{r}}_\alpha \cdot \dot{\boldsymbol{r}}_\alpha\right) \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial \dot{q}_i}\left(\frac{1}{2} m_\alpha \dot{\boldsymbol{r}}_\alpha \cdot \dot{\boldsymbol{r}}_\alpha\right)\right) - \frac{\partial}{\partial q_i}\left(\frac{1}{2} m_\alpha \dot{\boldsymbol{r}}_\alpha \cdot \dot{\boldsymbol{r}}_\alpha\right) \\ &= \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T_\alpha}{\partial \dot{q}_i} - \frac{\partial T_\alpha}{\partial q_i} \end{aligned}$$

so the D'Alembert's principle becomes

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} = Q_i \quad \text{for } i = 1, \ldots, n$$

and defining $T = \sum_\alpha T_\alpha$, $Q_i = \sum_\alpha \mathbf{F}_\alpha^{(a)} \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i}$, in this case, Q_i is called a generalized force. The previous equations are called Lagrange's equations.

We can now further decompose the applied force into a conservative term plus a nonconservative one $F_\alpha^{(a)} = \mathbf{F}_\alpha^{(\nabla)} + f_\alpha$, so

$$\mathbf{F}_\alpha^{(\nabla)} = -\nabla_\alpha V = -\frac{\partial V}{\partial \boldsymbol{r}_\alpha} = -\frac{\partial V}{\partial \boldsymbol{r}_{j,\alpha}} e_j$$

and

$$\mathbf{F}_\alpha^{(\nabla)} \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = -\frac{\partial V}{\partial \boldsymbol{r}_{j,\alpha}} e_j \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} e_l = -\frac{\partial V}{\partial \boldsymbol{r}_j}\frac{\partial \boldsymbol{r}_j}{\partial q_i} = -\frac{\partial V}{\partial q_i} = -\nabla_{q_i} V$$

so

$$Q_i = -\frac{\partial V}{\partial q_i} + Q_i^{(nc)}, \qquad Q_i^{(nc)} = f_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i}$$

hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} = -\frac{\partial V}{\partial q_i} + Q_i^{(nc)} \longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial}{\partial q_i}(T - V) = Q_i^{(nc)}$$

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Last class we proved that

$$\dot{p}_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T_\alpha}{\partial \dot{q}_i} - \frac{\partial T_\alpha}{\partial q_i}$$

so that on defining $T \equiv \sum_\alpha T_\alpha$, we get that D'Alembert's principle is equivalent to

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} = \sum_\alpha \mathbf{F}_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} \tag{$*$}$$

In particular, if $\mathbf{F}_\alpha = -\frac{\partial E_p}{\partial \boldsymbol{r}_{j,\alpha}} e_j \equiv -\frac{\partial E_p}{\partial \boldsymbol{r}_\alpha} \equiv -\nabla_\alpha E_p$, where $\boldsymbol{r}_{j,\alpha}$ is the j-th component of the α-th particle position vector, then

$$\sum_\alpha \mathbf{F}_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = -\frac{\partial V}{\partial \boldsymbol{r}_{j,\alpha}} e_j \cdot \frac{\partial \boldsymbol{r}_{\sigma,\alpha}}{\partial q_i} e_\sigma = -\frac{\partial V}{\partial q_i}$$

$$\longrightarrow V(q_i,t) = E_p(\boldsymbol{r}_1(q_i,t),\ldots,\boldsymbol{r}_N(q_i,t))$$

and $(*)$ transforms to

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i} = -\frac{\partial V}{\partial q_i} \equiv \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial (T-V)}{\partial q_i} = Q_i^{(nc)}$$

If $V \neq V(\dot{q}_i)$ or if $\sum_\alpha E_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial V}{\partial \dot{q}_i} - \frac{\partial V}{\partial q_i}$, then

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial (T-V)}{\partial \dot{q}_i} - \frac{\partial (T-V)}{\partial q_i} = Q_i^{(nc)}$$

If we write $L = T - V$, these are known as the Euler-Lagrange equations (standard form).

Notice that we can write the Euler-Lagrange equations as

$$\frac{\partial^2 L}{\partial \dot{q}_j \partial \dot{q}_i}\ddot{q}_j + \frac{\partial^2 L}{\partial q_j \partial \dot{q}_i}\dot{q}_j + \frac{\partial^2 L}{\partial t \partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0 \qquad (**)$$

So that if we define a matrix H by $(H)_{ij} = \frac{\partial^2 L}{\partial \dot{q}_j \partial \dot{q}_i}$, we see that if $\det H \neq 0$, then $(**)$ constitute a set of n 2nd-order differential equations. If such thing happens we say that L is a **regular** Lagrangian, otherwise it is called **singular**.

Before considering more usual examples, let us consider the case of velocity-dependent potentials with a very important example.

Consider a scalar function $V = V(q_i, \dot{q}_i, t)$ such that

$$Q_i \equiv \sum_\alpha E_\alpha \cdot \frac{\partial \boldsymbol{r}_\alpha}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial V}{\partial \dot{q}_i} - \frac{\partial V}{\partial q_i} = \frac{\partial^2 V}{\partial \dot{q}_j \partial \dot{q}_i}\ddot{q}_j + \frac{\partial^2 V}{\partial q_j \partial \dot{q}_i}\dot{q}_j + \frac{\partial^2 V}{\partial t \partial \dot{q}_i} - \frac{\partial V}{\partial q_i} \qquad (**)$$

if we further assume that $Q_i = Q_i(q_i, \dot{q}_i, t)$, then

$$\frac{\partial^2 V}{\partial \dot{q}_j \partial \dot{q}_i} = 0,$$

so that $V = \alpha_k \dot{q}_k + \beta$ with $\alpha_k = \alpha_k(q_i,t)$, $\beta = \beta(q_i,t)$ so $(**)$ now becomes

$$Q_i = -\frac{\partial(\alpha_k \dot{q}_k)}{\partial q_i} - \frac{\partial \beta}{\partial q_i} + \frac{\partial^2(\alpha_k \dot{q}_k)}{\partial t \partial \dot{q}_i} + \underbrace{\frac{\partial^2 \beta}{\partial t \partial \dot{q}_i}}_{=0} + \frac{\partial^2(\alpha_k \dot{q}_k)}{\partial q_j \partial \dot{q}_i}\dot{q}_j + \frac{\partial^2 \beta}{\partial q_j \partial \dot{q}_i}\dot{q}_j + \frac{\partial(\alpha_k \dot{q}_k)}{\partial \dot{q}_j \partial \dot{q}_i}\ddot{q}_j$$

$$= -\frac{\partial \alpha_k}{\partial q_i}\dot{q}_k - \frac{\partial \beta}{\partial q_i} + \frac{\partial \alpha_k}{\partial t}\delta_{ik} + \frac{\partial \alpha_k}{\partial q_j}\delta_{ik}\dot{q}_j + \frac{\partial \alpha_k}{\partial \dot{q}_j}\delta_{ik}\ddot{q}_j$$

$$= -\frac{\partial \alpha_k}{\partial q_i}\dot{q}_k - \frac{\partial \beta}{\partial q_i} + \frac{\partial \alpha_i}{\partial t} + \frac{\partial \alpha_i}{\partial q_j}\dot{q}_j + \underbrace{\frac{\partial \alpha_i}{\partial \dot{q}_j}}_{=0}\ddot{q}_j$$

$$= \dot{q}_j\left(-\frac{\partial \alpha_j}{\partial q_i} + \frac{\partial \alpha_i}{\partial q_j}\right) - \frac{\partial \beta}{\partial q_i} + \frac{\partial \alpha_i}{\partial t}$$

meaning that the Q_i must be functions of the first degree in the $\dot{q}_j's$.

Theorem: Given a set of functions $Q_i = Q_i(q_j, \dot{q}_j, t)$, there exist (locally) functions $\alpha_i(q_j, t)$ and $\beta(q_j, t)$ such that

$$Q_i = \left(\frac{\partial \alpha_i}{\partial q_j} - \frac{\partial \alpha_j}{\partial q_i}\right)\dot{q}_j - \frac{\partial \beta}{\partial q_i} + \frac{\partial \alpha_i}{\partial t}$$

hold if and only if the Q_i's satisfy

$$\frac{\partial^2 Q_i}{\partial \dot{q}_j \partial \dot{q}_k} = 0, \qquad \frac{\partial Q_i}{\partial \dot{q}_j} = -\frac{\partial Q_j}{\partial \dot{q}_i},$$

$$\frac{\partial^2 Q_i}{\partial q_k \partial \dot{q}_j} + \frac{\partial^2 Q_j}{\partial q_i \partial \dot{q}_k} + \frac{\partial^2 Q_k}{\partial q_j \partial \dot{q}_i} = 0, \qquad \frac{\partial Q_i}{\partial q_j} - \frac{\partial Q_j}{\partial q_i} = \frac{\partial^2 Q_i}{\partial t \partial \dot{q}_j} + \frac{\partial^2 Q_i}{\partial q_k \partial \dot{q}_j}\dot{q}_k. \qquad (***)$$

Example: For $n = 3$, the force on a charged particle (with charge q),

$$F = q(E + v \times B) \qquad E = E(\boldsymbol{r}, t),\ B = B(\boldsymbol{r}, t)$$

is given by the Lorentz equation. Using Cartesian coordinates, we will show that the components of the force satisfy conditions $(***)$ under some appropriate conditions. This is very interesting, because the "only" example of a force derivable from a potential as in $(**)$ is the electromagnetic force.

In Cartesian coordinates one has $\boldsymbol{r} = x\hat{i} + y\hat{j} + z\hat{k}$ so that $\partial \boldsymbol{r}/\partial q_i = \hat{i}$ for $q_i = x_i$. Hence $Q_i = F \cdot \hat{i} \equiv F_i$ (e.g. $Q_1 = F \cdot \hat{x} = F_x$) and thus

$$Q_i = q\,(E_i + \varepsilon_{iml} v_m B_l).$$

For the first equation in $(***)$ we have

$$\frac{\partial^2 Q_i}{\partial v_j \partial v_k} = 0.$$

For the second,

$$\frac{\partial Q_i}{\partial v_j} = q\varepsilon_{ijl} B_l \implies \frac{\partial Q_i}{\partial v_j} = -\frac{\partial Q_j}{\partial v_i}.$$

For the third one,

$$\frac{\partial^2 Q_i}{\partial x_k \partial v_j} = \frac{\partial}{\partial x_k}\left(q\varepsilon_{ijl} B_l\right) = q\varepsilon_{ijl}\frac{\partial B_l}{\partial x_k}$$

So

$$\frac{\partial^2 Q_j}{\partial x_i \partial v_k} = q\varepsilon_{jkl}\frac{\partial B_l}{\partial x_i} \quad \text{and} \quad \frac{\partial^2 Q_k}{\partial x_j \partial v_i} = q\varepsilon_{kil}\frac{\partial B_l}{\partial x_j},$$

where

$$\frac{\partial^2 Q_i}{\partial x_k \partial v_j} + \frac{\partial^2 Q_j}{\partial x_i \partial v_k} + \frac{\partial^2 Q_k}{\partial x_j \partial v_i} = q\left(\varepsilon_{ijl}\frac{\partial B_l}{\partial x_k} + \varepsilon_{jkl}\frac{\partial B_l}{\partial x_i} + \varepsilon_{kil}\frac{\partial B_l}{\partial x_j}\right). \qquad (1)$$

For the last equation we compute

$$\frac{\partial Q_i}{\partial x_j} = q\left(\frac{\partial E_i}{\partial x_j} + \varepsilon_{iml} v_m \frac{\partial B_l}{\partial x_j}\right)$$

so that

$$\frac{\partial Q_j}{\partial x_i} = q\left(\frac{\partial E_j}{\partial x_i} + \varepsilon_{jml} v_m \frac{\partial B_l}{\partial x_i}\right),$$

moreover,

$$\frac{\partial^2 Q_i}{\partial t \partial v_j} = \frac{\partial}{\partial t}\left(q\varepsilon_{ijl} B_l\right) = q\varepsilon_{ijl}\frac{\partial B_l}{\partial t}$$

and from before:

$$\frac{\partial^2 Q_i}{\partial x_k \partial v_j} = q\varepsilon_{ijl}\frac{\partial B_l}{\partial x_k}$$

so we get the condition

$$q\left(\frac{\partial E_i}{\partial x_j} + \varepsilon_{iml}v_m\frac{\partial B_l}{\partial x_j} - \frac{\partial E_j}{\partial x_i} - \varepsilon_{jml}v_m\frac{\partial B_l}{\partial x_i}\right) = q\left(\varepsilon_{ijl}\frac{\partial B_l}{\partial t} + \varepsilon_{ijl}\frac{\partial B_l}{\partial x_k}v_k\right)$$

We therefore conclude that in order for the four equations to hold, we require that

$$\varepsilon_{ijl}\frac{\partial B_l}{\partial x_k} + \varepsilon_{jkl}\frac{\partial B_l}{\partial x_i} + \varepsilon_{kil}\frac{\partial B_l}{\partial x_j} = 0 \quad \text{and} \quad \frac{\partial E_i}{\partial x_j} - \frac{\partial E_j}{\partial x_i} = \varepsilon_{ijl}\frac{\partial B_l}{\partial t} + v_k\left(\varepsilon_{ijl}\frac{\partial B_l}{\partial x_k} + \varepsilon_{jkl}\frac{\partial B_l}{\partial x_i} + \varepsilon_{kil}\frac{\partial B_l}{\partial x_j}\right)$$

Therefore, it is now clear that if any two indices are equal, the latter identically vanishes, so for all three indices distinct (recall we are in the case $n = 3$), we have w.l.o.g. $i = 1, j = 2, k = 3$ so the expression becomes

$$q\left(\frac{\partial B_3}{\partial x_3} + \frac{\partial B_1}{\partial x_1} + \frac{\partial B_2}{\partial x_2}\right) \equiv q\nabla\cdot\mathbf{B}$$

which vanishes identically if and only if $\nabla\cdot\mathbf{B} = 0$.

Therefore, it is clear that for $i = j$ this is identically true, and for $i \neq j$ we get all the possible combinations $(i = 1, j = 2), (i = 1, j = 3), (i = 2, j = 1), (i = 2, j = 3), (i = 3, j = 1), (i = 3, j = 2)$, but by inspection, one sees that the combinations (i, j) and (j, i) yield the same result, so we only need to expand $(i = 1, j = 2), (i = 1, j = 3), (i = 2, j = 3)$ and also, notice that we write this as

$$\frac{\partial E_i}{\partial x_j} - \frac{\partial E_j}{\partial x_i} = \varepsilon_{ijl}\frac{\partial B_l}{\partial t} + v_k\left(\varepsilon_{ijl}\frac{\partial B_l}{\partial x_k} + \varepsilon_{jkl}\frac{\partial B_l}{\partial x_i} - \varepsilon_{ikl}\frac{\partial B_l}{\partial x_j}\right)$$

$$= \varepsilon_{ijl}\frac{\partial B_l}{\partial t} + v_k\left(\varepsilon_{ijl}\frac{\partial B_l}{\partial x_k} + \varepsilon_{jkl}\frac{\partial B_l}{\partial x_i} + \varepsilon_{kil}\frac{\partial B_l}{\partial x_j}\right)$$

Therefore these two equations are equivalent to

$$\nabla\cdot\mathbf{B} = 0 \quad \text{and} \quad \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \qquad \text{(ME)}$$

Remark: We now know from electromagnetic theory that the Lorentz force indeed satisfies equations $(***)$. Hence there exist functions α_i, β as in the previous theorem satisfying

$$Q_i = \left(\frac{\partial \alpha_i}{\partial x_j} - \frac{\partial \alpha_j}{\partial x_i}\right)v_j - \frac{\partial \beta}{\partial x_i} + \frac{\partial \alpha_i}{\partial t} = q\left(E_i + \varepsilon_{iml}v_m B_l\right)$$

$$= q\left(-\frac{\partial \phi}{\partial x_i} - \frac{\partial A_i}{\partial t} + \varepsilon_{iml}v_m\varepsilon_{lrs}\frac{\partial A_s}{\partial x_r}\right)$$

Remark: From vector analysis, these equations are equivalent to the existence of functions $\mathbf{A}, \phi$ such that $\mathbf{B} = \nabla\times\mathbf{A}$, $\mathbf{E} = -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}$.

This way

$$\left(\frac{\partial \alpha_i}{\partial x_j} - \frac{\partial \alpha_j}{\partial x_i}\right)v_j - \frac{\partial \beta}{\partial x_i} + \frac{\partial \alpha_i}{\partial t} = q\left(-\frac{\partial \phi}{\partial x_i} - \frac{\partial A_i}{\partial t} + \varepsilon_{iml}\varepsilon_{lrs}v_m\frac{\partial A_s}{\partial x_r}\right)$$

$$= q\left(-\frac{\partial \phi}{\partial x_i} - \frac{\partial A_i}{\partial t} + (\delta_{ir}\delta_{ms} - \delta_{is}\delta_{mr})v_m\frac{\partial A_s}{\partial x_r}\right)$$

$$= q\left(-\frac{\partial \phi}{\partial x_i} - \frac{\partial A_i}{\partial t} + v_s\frac{\partial A_s}{\partial x_i} - v_r\frac{\partial A_i}{\partial x_r}\right)$$

$$= q\left(-\frac{\partial \phi}{\partial x_i} - \frac{\partial A_i}{\partial t} + v_j\left(\frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j}\right)\right)$$

thus $\alpha_i = -qA_i$, $\beta = q\phi$ and the generalized potential is

$$V(x_i, v_i, t) = \alpha_k v_k + \beta = -qA_k v_k + q\phi = -q(\mathbf{A}\cdot\mathbf{v} - \phi).$$

Therefore, the Lagrangian associated to the Lorentz force is

$$L = \frac{1}{2}mv^2 + q\mathbf{A}\cdot\mathbf{v} - q\phi.$$

It is left as an exercise to prove that the Euler-Lagrange equations yield the expression for the Lorentz force for the Lagrangian above.